Skin Cancer Classification Using Deep Learning

Introduction

Skin cancer is one of the most prevalent forms of cancer worldwide, and early detection plays a critical role in improving patient survival rates. Traditional diagnostic methods rely heavily on expert dermatologists, which can be time-consuming and prone to subjective interpretation. With the advancement of artificial intelligence, deep learning techniques—particularly Convolutional Neural Networks (CNNs)—have shown significant potential in analyzing medical images and assisting in disease diagnosis.

This project focuses on developing a deep learning-based model for automatic classification of skin lesion images using the HAM10000 dataset

2.Objective

The primary objectives of this project are:

- To develop a CNN-based model for multi-class classification of skin lesion images
- To analyze the effectiveness of deep learning in medical image classification
- To evaluate model performance using accuracy, confusion matrix, and classification metrics
- To investigate challenges such as class imbalance in real-world datasets

1. Dataset Description

The dataset used in this project is the HAM10000 (Human Against Machine with 10000 training images) dataset. It contains dermatoscopic images of skin lesions categorized into seven different classes.

A subset of 3000 images was selected to reduce computational complexity and enable faster training. Each image is associated with metadata such as lesion type and image ID.

A key characteristic of this dataset is **class imbalance**, where certain lesion types have significantly more samples than others. This imbalance affects model learning and classification performance.

4. Methodology

4.1 Data Preprocessing

The metadata file was used to generate file paths for all images. Missing or invalid image paths were removed to ensure data integrity. The categorical lesion labels were converted into numerical values for model training.

4.2 Data Visualization

Exploratory data analysis was performed to understand the dataset distribution. The class distribution plot revealed significant imbalance among classes. Additionally, sample images were visualized to observe variations in lesion appearance.

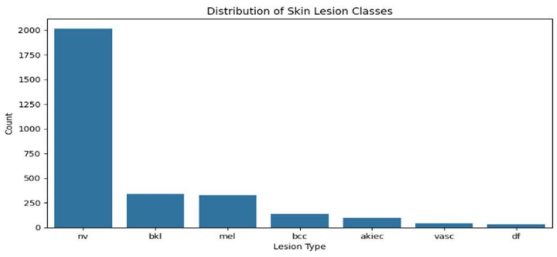

Figure 1: Class Distribution

4.3 Train-Test Split

The dataset was divided into training (80%) and testing (20%) sets using stratified sampling to preserve the class distribution in both subsets.

4.4 Data Augmentation

To improve model generalization and reduce overfitting, data augmentation techniques were applied. These include rotation, zooming, horizontal flipping, and normalization. Data augmentation helps the model learn invariant features from images.

4.5 Model Architecture

A Convolutional Neural Network (CNN) was designed to extract spatial features from images. The architecture consists of:

- Multiple convolutional layers with ReLU activation
- MaxPooling layers for dimensionality reduction
- Fully connected (dense) layers for classification
- Dropout layer to prevent overfitting
- Softmax output layer for multi-class classification

5. Model Training

The model was trained using the Adam optimizer and categorical cross-entropy loss function. Training was performed for 5 epochs with early stopping to prevent overfitting.

During training, both training and validation accuracy were monitored to evaluate learning performance.

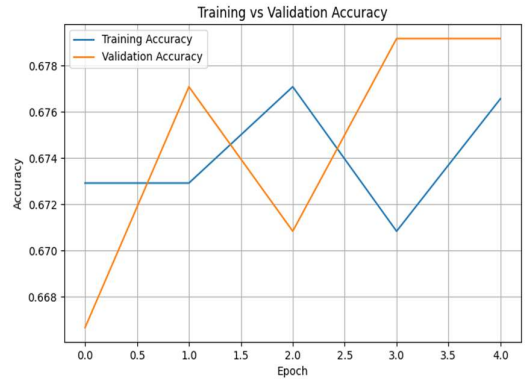

Figure 2: Training vs Validation Accuracy

6. Results and Evaluation

The model achieved the following performance:

Training Accuracy	68%
Validation Accuracy	68%
Test Accuracy	69%

The consistency between training and validation accuracy indicates that the model generalizes well without significant overfitting.

6.1 Confusion Matrix

The confusion matrix provides a detailed view of classification performance across different classes.

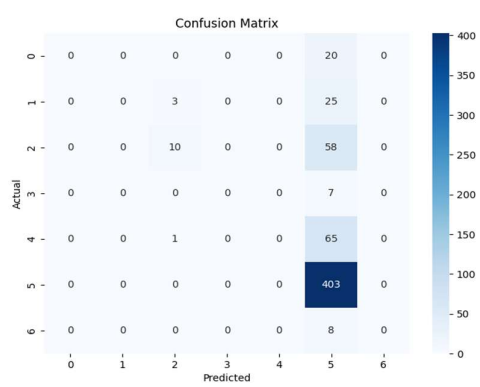

Figure 3: Confusion Matrix

6.2 Classification Analysis

The classification report shows that the model performs well for the majority class but poorly for minority classes. Some classes have very low precision and recall values, indicating that the model struggles to identify underrepresented categories.

7. Discussion

The results highlight an important real-world challenge in machine learning: **class imbalance**. The model tends to predict the dominant class more frequently, leading to high overall accuracy but poor performance for minority classes.

This demonstrates that accuracy alone is not sufficient for evaluating model performance, especially in medical applications where minority classes may represent critical conditions.

8. Conclusion

In this project, a CNN-based deep learning model was developed for skin cancer classification using the HAM10000 dataset. The model achieved an overall accuracy of approximately 69%, demonstrating the effectiveness of deep learning in image-based classification tasks.

However, the presence of class imbalance significantly impacts performance across different classes. Addressing this issue is essential for building reliable medical diagnostic systems.

9. Future Work

The performance of the model can be improved by:

Using transfer learning models such as ResNet, VGG16, or EfficientNet

- Applying class weighting or oversampling techniques
- Increasing dataset size and diversity
- Fine-tuning hyperparameters and network architecture
- Incorporating advanced augmentation techniques

These improvements can enhance model accuracy and ensure better performance across all classes.